Laplace transform:

$\mathcal{L}\{y'\}(s) = sY(s) - y(0)$

$\mathcal{L}\{y''\}(s) = s^2Y(s) - sy(0) - y'(0)$

$\mathcal{L}\{y'''\}(s) = s^3Y(s) - s^2y(0) - sy'(0) - y''(0)$

Convolution theorem:

$\mathcal{L}\{f*g\}(s) = \mathcal{L}\left\{\int_0^t f(t-\tau)g(\tau)\,d\tau\right\} = F(s)G(s)$

Some transforms:

$\mathcal{L}\{u(t-a)\} = e^{-as}\cdot\frac{1}{s}$

$\mathcal{L}\left\{\int_0^t g(v)\,dv\right\} = \frac{F(s)}{s}$

$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$

$\mathcal{L}\{e^{at}\sin(bt)\} = \frac{b}{(s-a)^2+b^2}$

$\mathcal{L}\{e^{at}\cos(bt)\} = \frac{(s-a)}{(s-a)^2+b^2}$

$\mathcal{L}\{t^n e^{at}\} = \frac{n!}{(s-a)^{n+1}}$, $n=1,2,3,\dots$

Interpolasjon:

$l_j = \prod_{\substack{m=0\\ m\neq j}}^{n} \frac{x-x_m}{x_j-x_m}$ $\quad L(x) = \sum_{j=0}^{n} l_j y_j$

Lagrange:

Newton:

$x_0\ y_0 = [y_0]$

$x_1\ y_1 = [y_1]$ $\quad [y_0,y_1]$

$x_2\ y_2 = [y_2]$ $\quad [y_1,y_2]$ $\quad [y_0,y_1,y_2]$

$x_3\ y_3 = [y_3]$ $\quad [y_2,y_3]$ $\quad [y_1,y_2,y_3]$ $\quad [y_0,y_1,y_2,y_3]$

$x_4\ y_4 = [y_4]$ $\quad [y_3,y_4]$ $\quad [y_2,y_3,y_4]$ $\quad [y_1,y_2,y_3,y_4]$ $\quad [y_0,y_1,y_2,y_3,y_4]$

$[y_0,y_1] = \frac{y_1-y_0}{x_1-x_0}$ $\qquad [y_0,y_1,y_2] = \frac{[y_1,y_2]-[y_0,y_1]}{x_2-x_0}$

$N(x) = [y_0] + [y_0,y_1](x-x_0) + [y_1,y_2,y_3](x-x_1)(x-x_0) + \cdots$

Fourier Analyse:

Periode $T = 2L$

$f(x) = \sum_{n=-\infty}^{\infty} C_n \exp\left(i\frac{n\pi x}{L}\right)$

$f(x) = a_0 + \sum_{n=1}^{\infty}\left[a_n\cos\left(\frac{n\pi x}{L}\right) + b_n\sin\left(\frac{n\pi x}{L}\right)\right]$

$C_n = \begin{cases} \frac{1}{2}(a_n - ib_n), & n>0 \\ \frac{1}{2}(a_n + ib_n), & n<0 \end{cases}$ $\qquad C_n = \frac{1}{2L}\int_{-L}^{L} f(x)\,e^{-i\frac{n\pi x}{L}}\,dx$

$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx = C_0$

$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$

$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$

Funksjoner:

$f(-x) = -f(x)$ odd: $f(x)$

$f(-x) = f(x)$ even: $g(x)$

Odd · odd = even

Even · even = even

Even · odd = odd

Odd extension $\to B_n$

Even extension $\to A_0, A_n$

Non-periodic functions Fourier extensions:

Eks: $f(x) = \sin(x)$ $\quad \int_{-L}^{L} f(x) = 0$

Eks: $g(x) = \cos(x), |x|$ Even $A_0 = \frac{1}{L}\int_0^L f(x)\,dx$, $A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi}{L}x\right)dx$ $\quad \int_{-L}^{L} g(x) = 2\int_0^L g(x)$

Odd: $B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi}{L}x\right)dx$

Fourier transform:

$\mathcal{F}\{f*g\}(\omega) = \sqrt{2\pi}\,[\mathcal{F}\{f\}\cdot\mathcal{F}\{g\}]$

$\mathcal{F}\{u'\}(\omega) = i\omega\,\mathcal{F}\{u\}(\omega)$

$\mathcal{F}\{u''\}(\omega) = -\omega^2\,\mathcal{F}\{u\}(\omega)$

PDE (FT):

Eks. $u_t = u_{xx} + u$, $u(x,0) = \begin{cases}\frac{\sin(x)}{x}, & x\neq 0\\ 1, & x=0\end{cases}$

- Fourier transformer begge sider mhp. x. $(\hat{u}(\omega,t))$
- Gjør om til ODE ved bruk av $\mathcal{F}\{u'\}\left(\frac{d\hat{u}(\omega,t)}{dt} = \hat{u}(\omega,t)(1+i\omega)\right)$
- Løs ODE. $(\hat{u}(\omega,t) = C(\omega)e^{(1+i\omega)t})$
- Transformer initial bet. for $C(\omega)$. $\left(C(\omega) = \hat{u}(\omega,0) = \begin{cases}\sqrt{\frac{2}{\pi}}, & |\omega|<1\\ 0, & |\omega|>1\end{cases}\right)$
- Sett inn. $\left(\hat{u}(\omega,t) = \begin{cases}\sqrt{\frac{2}{\pi}}e^{(1+i\omega)t}, & |\omega|<1\\ 0, & |\omega|>1\end{cases}\right)$ og invers transformer

Parsevals identitet:

$\int_{-L}^{L}(f(x))^2$ eksisterer og at $f(x) = \sum_{n=-\infty}^{\infty} C_n e^{i\frac{n\pi x}{L}}$

$\frac{1}{2L}\int_{-L}^{L}(f(x))^2\,dx = \sum_{n=-\infty}^{\infty}|C_n|^2 = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2+b_n^2)$

$E_N = \int_{-L}^{L}(f(x)-F_N(x))^2\,dx = \int_{-L}^{L} f(x)^2\,dx - L(2a_0^2 + \sum_{n=1}^{N}(a_n^2+b_n^2))$

Numerisk integrasjon:

$\int_a^b f(x)\,dx \approx \sum_{i=0}^{n} f(x_i)w_i = Q[f]$, ukjent Fors: $\int_a^b P_n(x)\,dx = \sum_{i=0}^{n} f(x_i)\int_a^b l_i(x)\,dx = \sum_{i=0}^{n} f(x_i)w_i$

Fra a til b. $Q[f](a,b)$

Midtpunkts: $QM[f] = (b-a)f\left(\frac{a+b}{2}\right)$

Trapesmetode: $QT[f] = (b-a)\left(\frac{1}{2}f(a) + \frac{1}{2}f(b)\right)$ $\Rightarrow d=1$

Simpsonsmetode: $QS[f] = \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right)$ $\Rightarrow d=3$

Simpsonsmetode krever at b-a er et partallmultippel av h. Dvs. $h<\varepsilon$, $h =$ når $\frac{b-a}{n} < \varepsilon$, $n=2,4,\dots$

Error: $h = \frac{b-a}{n}$

$|E_{QM}| = \frac{(b-a)}{6}\cdot h^2 \max_{x\in[a,b]}|f''(x)|$

$|E_{QT}| = \frac{(b-a)}{12}\cdot h^2 \max_{x\in[a,b]}|f''(x)|$

$|E_{QS}| = \frac{(b-a)}{180}\cdot h^4 \max_{x\in[a,b]}|f^{(4)}(x)|$

Presisjon d: $\int_a^b x^d\,dx = Q[x^d](a,b)$ for $k=1,2,\dots,d$

D'Alembert:

Bølge likning: $u_{tt} = c^2 u_{xx}$

Init.: $u(x,0) = f(x)$, $u_t(x,0) = g(x)$

$u(x,t) = \frac{1}{2}\left[f(x+ct) + f(x-ct) + \frac{1}{c}\int_{x-ct}^{x+ct} g(\tau)\,d\tau\right]$

Rounding errors:

Eks: $\varepsilon = 10^{-16}$, $\underbrace{O(h^2)}_{\text{approx. } \varepsilon} = O\left(\frac{\varepsilon}{h^2}\right)$ (rounding error / Denominator) $\Rightarrow h^4 = O(\varepsilon)$

$h = O(\varepsilon^{1/4}) = O(10^{-4})$

Newtons metode:

$x_{K+1} = x_K - \frac{f(x)}{f'(x)}$

- Funksjonen må være glatt slik at den deriverte oppfører seg normalt i nærheten av roten og ikke er lik null for å unngå å dele på null.

Kvadratisk konvergens, $P=2$:

$|x_{K+1} - r| \leq M|x_K - r|^2$

(Avhengig av oppgaven)

Flere dimensjoner:

$f_n(x_1, x_2, \dots, x_n)$ $\quad J_{i,j} = \begin{bmatrix}\frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n}\\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n}\end{bmatrix}$

$x_{K+1} = x_K - J^{-1}\vec{f}(x_K)$

- Hvor J er jacobi matrisen og x_K er en $1\times n$ vektor.

$\Rightarrow J\,\Delta K = -\vec{f}(x_K)$, $\Delta K = x_{K+1} - x_K$

1) Finn J og $\vec{f}(x_K)$

2) Løs for ΔK og sett inn i $x_{K+1} = x_K + \Delta K$

Bisection metoden:

- To punkter a,b slik at $f(a)f(b) < 0$
- Finn så midtpunktet $c = \frac{a+b}{2}$
- Sjekk om $f(c) = 0$, hvis ja er roten funnet.

Hvis ikke, fortsetter metoden.

- $f(a)f(c) < 0$ er roten på intervallet (a,c)
- Sett derfor $a_1 = a_0$, $b_1 = c_0$, $c_1 = \frac{a_1+b_1}{2}$
- $f(a)f(c) > 0 \Rightarrow$ roten $\in (c,d)$

Iterations: $k \gg \frac{\log\left(\frac{b-a}{2\cdot tol}\right)}{\log(2)}$

Fixed point:

$x = g(x)$

Anta at $g(x)$ er kontinuerlig på intervallet $[a,b]$. Anta også at $g'(x)$ eksisterer og at det finnes en konstant $L<1$.

(1) $a < g(x) < b \ \forall\ x\in[a,b]$ $\qquad$ (2) $|g'(x)| \leq L < 1 \ \forall\ x\in[a,b]$

- Hvis (1) holder, eksisterer det minst et fikspunkt på intervallet $[a,b]$
- Hvis (1) og (2) holder, er dette fikspunktet unikt og iterasjonen vil konvergere til dette fikspunktet for alle $x_0\in[a,b]$

$x_{K+1} = g(x_K)$

Error: $|x_{K+1} - r| \leq \frac{L^{K+1}}{1-L}|g(x_0) - x_0|$

$\varepsilon_{K+1} < tol$ minst $k \gg \frac{\ln\left(\frac{(1-L)\cdot tol}{|g(x_0)-x_0|}\right)}{\ln(L)} - 1$

FDM:

(Forward/Backwards gir ikke feilgrense pga. $\text{eps}/10^n$)

Forward: $u'(x) = \frac{u(x+h)-u(x)}{h} + O(h)$

Backward: $u'(x) = \frac{u(x)-u(x-h)}{h} + O(h)$

Central: $u'(x) = \frac{u(x+h)-u(x-h)}{2h} + O(h^2)$

$u''(x) = \frac{u(x+h)-2u(x)+u(x-h)}{h^2} + O(h^2)$

Taylor ekspansjon:

$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$ $\qquad u(x-h) = \sum_{n=0}^{K-1}\frac{u^{(n)}(x)}{n!}(-h)^n + O(h^K)$

Forward Euler (varme): (Eksplisitt) $u_t = u_{xx}$, $u(x,0) = f(x)$, $u(0,t) = u(L,t) = C$

$u_t = \frac{u(x,t+K)-u(x,t)}{K} + O(K)$ $\qquad u_{xx} = \frac{u(x+h,t)-2u(x,t)+u(x-h,t)}{h^2} + O(h^2)$

Scheme: $\frac{u_m^{n+1}-u_m^n}{K} = \frac{u_{m+1}^n - 2u_m^n + u_{m-1}^n}{h^2}$

Konvergensrate:

$p = \frac{\ln\left(\frac{e_{K+1}}{e_{K+2}}\right)}{\ln\left(\frac{e_K}{e_{K+1}}\right)}$

Konvergensrate p hvis: $e_{K+1} \leq C e_K^p$, $C>0$

Differensialligninger:

$ay'' + by' + cy = 0 \Rightarrow ar^2 + br + c = 0$

- En løsning: $y = (A+Bx)e^{r_1 x}$
- To løsning: $y = Ae^{r_1 x} + Be^{r_2 x}$
- Ingen løsning: $r = \Delta \pm ti$ $\quad y = e^{\Delta x}(A\sin(tx) + B\cos(tx))$

PDEs:

Lineær $\to$ alle u og dens deriverte er av grad 1.

Homogen $\to$ alle ledd avhenger av u

Separation of variables:

1) sett $u(x,t) = F(x)G(t)$ og finn ODE for F og G.

2) Finn alle ikke-trivielle løsning $u_n(x,t) = \sum_{n=1}^{\infty} F_n(x)G_n(t)$ (Superposisjon)

3) Finn alle ko-eff. som tilfredsstiller initial betingelsen.

Runge-kutta metoder:

A Runge-Kutta method with Butcher tableau (I) to solve the ODE $y' = f(t,y)$ is defined as (II)

I

c_1	$a_{11}\ \cdots\ a_{1s}$	
$\vdots$	$\vdots\ \ddots\ \vdots$	
c_s	$a_{s1}\ \cdots\ a_{ss}$	
	$b_1\ \cdots\ b_s$	

II

$k_1 := f(t_n + c_1 h,\ y_n + h\sum_{j=1}^{s} a_{1j}k_j)$

$k_s := f(t_n + c_s h,\ y_n + h\sum_{j=1}^{s} a_{sj}k_j)$

$y_{n+1} := y_n + h\sum_{j=1}^{s} b_j k_j$

Nyttig:

$e^{i\theta} = \cos\theta + i\sin\theta$ $\qquad \sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$ $\qquad \cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$

$\int u\cdot v' = u\cdot v - \int u'\cdot v$

$\int_{-\pi}^{\pi} e^{i(n-m)x}\,dx = \begin{cases}2\pi, & n=m\\ 0, & n\neq m\end{cases}$

$u(t-a) = \begin{cases}1, & t>a\\ 0, & t<a\end{cases}$

$\delta(t-a) = \begin{cases}\infty, & t=a\\ 0, & t\neq a\end{cases}$

$\sin\left(\frac{n\pi}{2}\right) = \sin\left(\frac{(2k+1)\pi}{2}\right) = (-1)^k \Rightarrow \frac{\sin\left(\frac{n\pi}{2}\right)}{n\pi} = \frac{(-1)^k}{(2k+1)\pi}$

$\cos\left(\frac{n\pi}{2}\right) = \cos\left(\frac{(2k)\pi}{2}\right) = (-1)^k \Rightarrow \frac{\cos\left(\frac{n\pi}{2}\right)}{n\pi} = \frac{(-1)^k}{(2k)\pi}$

$\int_{-\infty}^{\infty} f(t)\delta(t-a) = f(a)$

$f(t) * \delta(t-a) = f(t-a)$